the wave function in the **L**-basis. If we use the orthonormality properties of the basis vectors,

$$\langle \lambda_i | \lambda_j \rangle = \delta_{ij},$$

then the wave function in the **L**-basis may also be identified with the inner products (or projections) of the state-vector |Ψ> onto the eigenvectors |λ>:

$$\psi(\lambda) = \langle \lambda | \Psi \rangle.$$

You can think of the wave function in two ways. First of all, it is the set of components of the state-vector in a particular basis. These components can be stacked up to form a column vector:

$$\begin{pmatrix} \psi(\lambda_1) \\ \psi(\lambda_2) \\ \psi(\lambda_3) \\ \psi(\lambda_4) \\ \psi(\lambda_5) \end{pmatrix}.$$

Another way to think of the wave function is as a function of λ. If you specify any allowable value of λ, the function $\psi(\lambda)$ produces a complex number. One can therefore say that

$$\psi(\lambda)$$

is a *complex-valued function of the discrete variable* λ. When thought of in this way, linear operators become operations that are applied to functions, and give back new functions.

One last reminder: the probability for an experiment to have outcome λ is